New Keynesian Model

Giovanni Pio Cirillo

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Introduction

New Keynesian Models (NKM) represent an important evolution in the field of macroeconomics, developed starting from the mid-1980s in response to the limitations of Real Business Cycle (RBC) models. While RBC models are based on the idea that economic fluctuations are primarily the result of technological shocks and that markets are perfectly efficient, New Keynesian models introduce some key hypotheses that seek to resolve certain criticalities of that approach, such as:

- **Non-neutrality of money:** shocks can be not only of a real nature but also of a monetary nature, whereby even the behavior of the central bank can generate economic fluctuations;
- **Sticky prices and wages:** the economy is characterized by real rigidities, due to the presence of intermediate firms operating under monopolistic competition, and by nominal rigidities, due to the inability (or unwillingness) of a certain fraction of firms to optimize prices following a shock. Furthermore, since wages are sticky, NKM models allow for the presence of involuntary unemployment;
- **Endogenous persistence of shocks:** given the hypothesis of a closed economy without capital accumulation and characterized by the presence of real and nominal frictions, in NKM models, the persistence of shocks is determined endogenously.

To better understand the macroeconomic dynamics that emerge from the application of the hypotheses just introduced, a simple NKM model is developed below, the equilibrium of which is determined by the optimal choices of four representative economic agents: households, intermediate firms, final firms, and the central bank.

Households

Households must solve two maximization problems: intratemporal maximization and intertemporal maximization. The intratemporal maximization problem concerns the optimal choice of the good to consume at each instant of time t among those available in the basket. The consumption of the j-th good within the basket depends on its relative price with reference to the aggregate price index, namely:

$$c_t(j) = \frac{P_t(j)^{-\eta}}{P_t} \cdot c_t \tag{1}$$

where:

- $P_t(j)$ is the price of the j-th good at time t;
- P_t is the aggregate price index;
- η is a parameter that measures the elasticity of substitution between two different goods;
- c_t is the aggregate consumption.

Therefore, as the price of the j-th good increases, all other conditions being equal, its demand by households reduces.

The intertemporal maximization problem concerns the optimal choice of consumption between two successive periods of time. The New Keynesian household wants to maximize its intertemporal utility, but must respect a budget constraint that imposes equality between income and expenditure for each period. Thus, it faces the following constrained maximization problem:

$$\max_{c_t, L_t, Q_{t+1}} U = E_0 \sum_{t=0}^{\infty} \beta^t \left[\log(c_t) - \frac{(L_t)^{1+\gamma}}{1+\gamma} \right] \tag{2}$$

subject to:

$$c_t + \frac{1}{R_t} \frac{B_t}{P_t} + E_t \zeta_{t,t+1} \frac{Q_{t+1}}{P_t} = \frac{B_{t-1}}{P_t} + \frac{Q_t}{P_t} - T_t + w_t L_t \tag{3}$$

Households, therefore, have a utility function that is increasing in consumption and decreasing in labor supply. The parameter β indicates the intertemporal discount factor (i.e., how much households value future consumption in current consumption choices) while the parameter γ corresponds to the inverse of the Frisch Elasticity (i.e., the propensity of households to supply labor as the wage varies).

The budget constraint expressed in real terms imposes equality between the expenditures and income of the representative household. Expenditures are due to:

- consumption c_t;
- purchase of riskless bonds with nominal value B_t which will be repaid in the next period at the interest rate R_t;
- purchase of assets ("nominal state contingent assets") that will mature in the next period Q_{t+1} and whose current price is determined by the expected future value $E_t \zeta_{t,t+1}$.

Income, on the other hand, originates from:

- repayment of riskless bonds purchased in the previous period B_{t-1};
- return on assets purchased in the previous period Q_t;
- transfer from households to the government in the form of taxes or from the government to households in the form of subsidies ("lump sum" transfer) of real amount T_t;
- labor income $w_t L_t$.

The problem of maximizing the intertemporal utility of the New Keynesian household can be solved by applying the Lagrangian and developing it over two periods (t and $t+1$), obtaining:

$$\begin{aligned}\mathcal{L}_t^H &= [\log(c_t) - \frac{(L_t)^{1+\gamma}}{1+\gamma}] + \beta[\log(c_{t+1}) - \frac{(L_{t+1})^{1+\gamma}}{1+\gamma}] \\ &+ \lambda_t[\frac{B_{t-1}}{P_t} + \frac{Q_t}{P_t} - T_t + w_t L_t - c_t - \frac{1}{R_t}\frac{B_t}{P_t} - E_t\zeta_{t,t+1}\frac{Q_{t+1}}{P_t}] \\ &+ \beta\lambda_{t+1}[\frac{B_t}{P_{t+1}} + \frac{Q_{t+1}}{P_{t+1}} - T_{t+1} + w_{t+1}L_{t+1} - c_{t+1} - \frac{1}{R_{t+1}}\frac{B_{t+1}}{P_{t+1}} - E_{t+1}\tilde{\zeta}_{t+1,t+2}\frac{Q_{t+2}}{P_{t+1}}]\end{aligned}$$

The choice variables on which the household can act are consumption c_t, labor supply L_t, and the demand for assets Q_{t+1}, for which we have the following first-order conditions (FOCs):

- $\frac{\partial\mathcal{L}}{\partial c_t} = 0 \rightarrow \lambda_t = c_t^{-1} = \frac{1}{c_t}$ (NKM Eq. 1 - Marginal Utility of Consumption)
- $\frac{\partial\mathcal{L}}{\partial L_t} = 0 \rightarrow w_t = \frac{L_t^\gamma}{\lambda_t}$ (NKM Eq. 2 - Marginal Utility of Labor or optimal labor supply)
- $\frac{\partial\mathcal{L}}{\partial Q_{t+1}} = 0 \rightarrow \lambda_t = (\frac{R_t}{\pi_{t+1}})\beta\lambda_{t+1}$ (NKM Eq. 3 - Euler Equation)

Note: In performing the partial derivative $\frac{\partial\mathcal{L}}{\partial Q_{t+1}} = 0$, the following substitutions were made: $\frac{P_t}{P_{t+1}} = \frac{1}{\frac{P_{t+1}}{P_t}} = \frac{1}{\pi_{t+1}}$ where π_{t+1} is the expected inflation in the next period; $E_t\zeta_{t,t+1} = \frac{1}{R_t}$, which applies the no-arbitrage condition between riskless bonds and assets.

Intermediate Firms

Intermediate firms operate under monopolistic competition and are therefore the main source of real rigidity in NKM models. Intermediate firms produce a homogeneous good by combining technology and labor according to the following production function:

$$Y_t = A_t L_t \quad \text{(NKM Eq. 4 - Production function)} \tag{4}$$

As hypothesized by RBC theory, the level of technology A_t present in the production function follows a law of motion expressed by the following autoregressive process:

$$\log(A_t) = \rho\log(A_{t-1}) + \epsilon_t \quad \text{(NKM Eq. 5 - TFP law of motion)} \tag{5}$$

where:

- $\rho\log(A_{t-1})$ is the deterministic component whose weight is established by the parameter ρ, which indicates the persistence of the TFP shock ($0 < \rho < 1$);
- ϵ_t is the stochastic component, from which the actual shock originates.

Intermediate firms face a problem of minimizing production costs (related to labor compensation) subject to the production constraint, which can be specified as follows:

$$\min_{L_t} w_t L_t \quad \text{s.t. } Y_t = A_t L_t \tag{6}$$

To solve the minimization problem, the Lagrangian can be applied, defining the Lagrange multiplier as mc_t to simplify its identification:

$$\mathcal{L} = w_t L_t + mc_t(Y_t - A_t L_t) \tag{7}$$

The choice variable of the intermediate firm is the demand for labor L_t, for which the first-order condition is as follows:

$$\frac{\partial \mathcal{L}}{\partial L_t} = 0 \rightarrow w_t = mc_t A_t \quad \text{(NKM Eq. 6 - Optimal labor demand)} \tag{8}$$

Defined this way, the term mc_t indicates the marginal cost of labor. For simplicity, assume that in the economic system under examination, intermediate firms also perform the role of final firms; thus, it is they who place the produced goods on the market at an optimal price.

Price Setting and Calvo Rule

In particular, we observe the presence of a certain fraction ω of firms that, when a shock occurs, fail to optimize prices, while the complementary fraction of firms $(1-\omega)$ manages to update the prices charged on the market. The parameter ω (stickiness parameter) can also be interpreted as the probability that a generic firm in each period t is characterized by sticky prices: this hypothesis is the source of nominal rigidities in the NKM model.

To determine the aggregate price index in the economy, a simple Calvo rule (Calvo, 1983) can be introduced, whereby firms that fail to optimize prices at time t will apply the aggregate price of the previous period $t-1$; therefore, the aggregate price at time t is given by:

$$P_t = (1-\omega) \cdot P_t^* + \omega \cdot P_{t-1} \tag{9}$$

The intermediate firm produces a single homogeneous good that it sells on the market at price $P_t(j)$. Therefore, assuming it also acts as a final firm, in each period t it must face a dynamic problem of constrained profit maximization:

$$\max_{P_t(j)} \Pi = E_t \sum_{i=0}^{\infty} \omega^i \xi_i \left[\frac{P_t(j)}{P_{t+i}} Y_{t+i}(j) - mc_{t+i} Y_{t+i}(j) \right] \tag{10}$$

subject to:

$$Y_t(j) = c_t(j) = \frac{P_t(j)^{-\eta}}{P_t} \cdot c_t \tag{11}$$

where:

- ω^i is the already introduced stickiness parameter. Note that as ω increases, the probability that the firm at time i fails to optimize prices increases, so it assigns greater value to future profits compared to a firm with flexible prices;
- ξ_i is the ratio between the marginal utilities of consumption between two successive periods of time, namely $\xi_i = \beta^t \frac{UMC_{t+1}}{UMC_t} = \beta^t \frac{\lambda_{t+1}}{\lambda_t}$.

Note that the constraint of the final firm is nothing other than the equality between the supply of goods and the demand for goods (or rather, of the j-th good) by households. From the resolution of the constrained maximization problem by the final firm, a New

Keynesian Phillips Curve (Phillips, 1958) is obtained, which in this linearized version links current inflation to the weight of marginal production costs, to the discount factor of households, and to expected future inflation. Therefore, the NKM model in question features forward-looking inflation:

$$\hat{\pi}_t = k \cdot \hat{mc}_t + \beta \cdot E_t[\hat{\pi}_{t+1}] \quad \text{(NKM Eq. 7 - Linearized NKPC)} \tag{12}$$

with $k = \frac{(1-\omega)(1-\beta\omega)}{\omega}$.

Central Bank and Monetary Policy

Finally, the central bank pursues its monetary policy through the instrument of the nominal interest rate R_t, which it sets according to the following simplified Taylor rule (Taylor, 1993):

$$\frac{R_t}{R} = \left(\frac{\pi_t}{\pi}\right)^{\varphi_\pi} + is_t \quad \text{(NKM Eq. 8 - Taylor rule)} \tag{13}$$

where:

- R is the central bank's target interest rate;
- π is the target inflation level;
- φ_π is the so-called "Taylor parameter", which summarizes the weight assigned by the central bank to the goal of inflation stability in its mandate. According to the Taylor principle, the value of the Taylor parameter should be greater than one ($\varphi_\pi > 1$), as it is appropriate for the central bank to respond more than proportionally to deviations of inflation from its target level;
- is_t is the endogenous variable from which the monetary shock originates.

In particular, the monetary shock follows an autoregressive process defined as follows:

$$is_t = \rho_m \cdot is_{t-1} + ms_t \quad \text{(NKM Eq. 9 - Monetary shock law of motion)} \tag{14}$$

Finally, it is necessary to close the NKM model with a market clearing condition:

$$Y_t = c_t \quad \text{(NKM Eq. 10 - Market clearing condition)} \tag{15}$$

The NKM model just constructed consists of ten endogenous variables, distributed as follows:

- technology A and monetary shock is are state variables, as they are explicit at both time $t-1$ and time t;
- wage W, labor L, output Y, consumption C, nominal interest rate R, and marginal labor cost mc are static variables as they appear only at time t;
- marginal utility of consumption λ and inflation π are jump variables because they guide the dynamics from time t to time $t+1$.

The model also features six parameters $(\gamma, \beta, k, \varphi_\pi, \rho, \rho_m)$ whose values are assigned through calibration or by drawing from the reference scientific literature. Once the system of equations describing the NKM model has been introduced, it is possible to proceed with its linearization around its steady state, in order to make the observation of the economy's behavior easier.

From the resolution of the steady state, we have:

$$
\begin{aligned}
\hat{w}_t &= \gamma \cdot \hat{l}_t - \hat{\lambda}_t \\
\hat{\lambda}_t &= \hat{r}_t - E_t\hat{\pi}_{t+1} + E_t\hat{\lambda}_{t+1} \\
\hat{y}_t &= \hat{a}_t + \hat{l}_t \\
\hat{w}_t &= \hat{mc}_t + \hat{a}_t \\
\hat{a}_t &= \rho \cdot \hat{a}_{t-1} + \epsilon_t \\
\hat{\pi}_t &= k \cdot \hat{mc}_t + \beta \cdot E_t[\hat{\pi}_{t+1}] \\
\hat{r}_t &= \varphi_\pi \cdot \hat{\pi}_t + \hat{is}_t \\
\hat{is}_t &= \rho_m \cdot \hat{is}_{t-1} + ms_t \\
\hat{y}_t &= \hat{c}_t
\end{aligned}
$$

Note 2: By convention, steady-state endogenous variables are indicated with the lowercase letter of the corresponding level variable (e.g., a instead of A).

Simulation and Results

Starting from a steady-state condition, we want to observe the dynamics of the NKM model in order to understand the trend of economic variables in response to a real TFP shock and a nominal monetary shock. Simulations are performed using Dynare software.

TFP Shock Analysis

Assume first that a positive TFP shock occurs, leading to an improvement in technology. It can be noted that, following the technology shock, the real wage $\hat{w}_t$ increases but to a much more limited extent than what occurred in the RBC model. Therefore, it can be stated that the NKM model manages to resolve, or at least limit, the productivity puzzle.

Furthermore, observing the consequences of the shock on the entire economic system, the following dynamics emerge: the improvement in technology ($\hat{a}_t > 0$) determines an increase in the marginal productivity of factors (i.e., labor), whereby, ceteris paribus, a reduction in the marginal cost of labor occurs ($\hat{mc}_t < 0$). As production costs reduce, firms are pushed to adjust prices downward, so inflation reduces ($\hat{\pi}_t < 0$) and the central bank, in an attempt to bring inflation back to equilibrium, reduces the nominal interest rate ($\hat{r}_t < 0$). The lower interest rate reduces the marginal utility of future consumption ($\hat{\lambda}_t < 0$), thereby discouraging investment and pushing households to consume more ($\hat{c}_t > 0$).

However, given the hypothesis of nominal rigidities introduced with the Calvo rule, there is a certain fraction ω of firms with sticky prices that charge a higher price than that which would allow household consumption to fully absorb the supply of goods. Therefore, in order to guarantee equality between demand and supply in the goods market, firms

decide to reduce the demand for labor ($\hat{l}_t < 0$). The reduction in labor demand - that is, the increase in unemployment following a TFP shock with a simultaneous increase in aggregate production - is called the "employment issue".

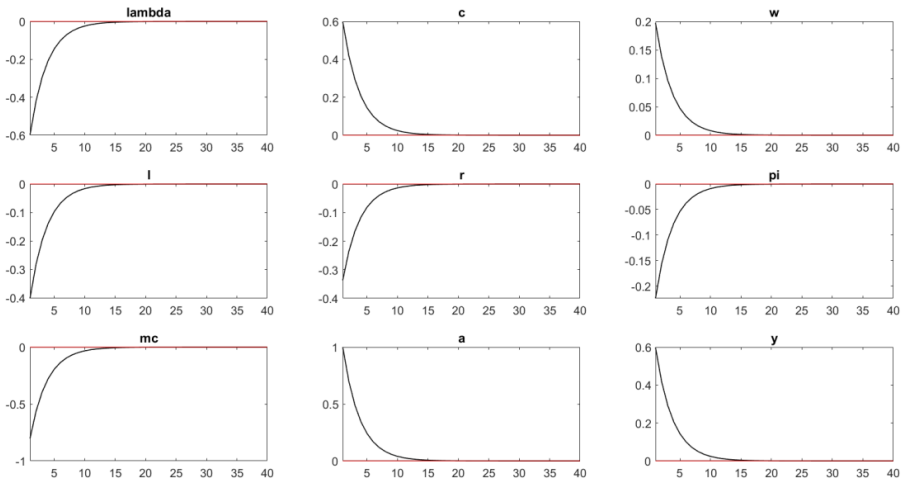

Figure 1: TFP Shock in a NK Model

Monetary Shock Analysis

Assume, then, that a negative monetary shock occurs, due for example to the central bank's response to the growth of inflation driven by production costs. The restrictive monetary policy determines an increase in the nominal interest rate ($\hat{r}_t > 0$), which discourages consumption by households in favor of investment ($\hat{c}_t < 0$). The reduction in consumption results in lower demand for goods, so firms are pushed to reduce aggregate supply by employing a smaller amount of labor ($\hat{l}_t < 0$). The growth of unemployment pushes households to accept lower wages ($\hat{w}_t < 0$), resulting in a reduction in the marginal cost of production ($\hat{mc}_t < 0$). Lower production costs are absorbed by a reduction in prices by firms - or at least by those with flexible prices. This ultimately allows the central bank to pursue its objective of reducing inflation ($\hat{\pi}_t < 0$), albeit at the cost of lower output and higher unemployment.

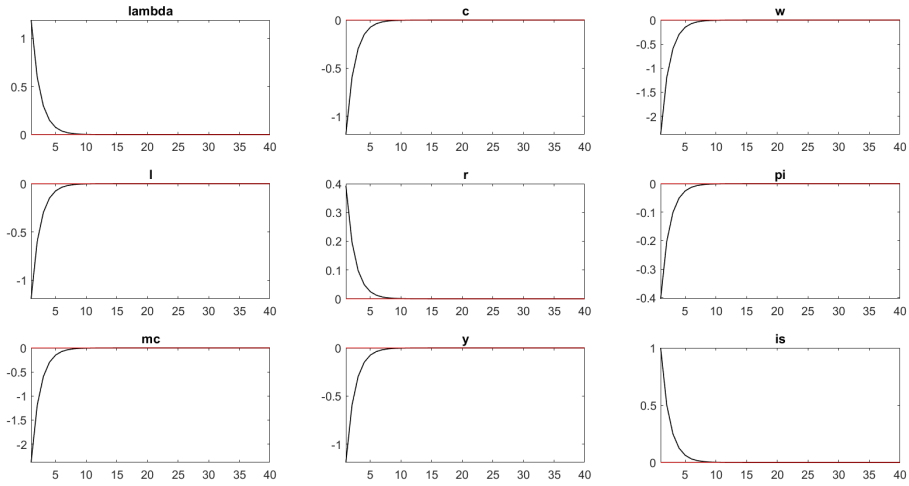

Figure 2: Monetary Shock in a NK Model